

Year 7 · Chapter 6 · Geometrical figures: answers (teacher copy)

6.01 Transforming shapes (booklet p. 3)

Examples (model these live)

E1 Translate the shape 4 right and 2 down. Draw the image.
move every corner 4 right, 2 down · join the new corners

E2 Reflect the shape in the dashed line.
each corner the same distance on the other side · join the new corners

E3 Rotate the shape 90° clockwise about the dot.
turn each corner a quarter turn · join the new corners

Questions (26 on today's skill, 10 from earlier lessons)

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|---|---|---|---|
| 1 translation | 14 mirror image: each corner the same distance from the line, on the other side | 22 a quarter turn clockwise about the dot (check each corner) | 30 90° anticlockwise (or 270° clockwise). |
| 2 7 right, 3 up | | 23 a half turn: each corner the same distance on the opposite side of the dot | 31 20 4.07 |
| 3 reflection | | 24 x 5.09 | 32 a quarter turn clockwise about the dot (check each corner) |
| 4 reflection | | 25 A half turn (180°). | 33 $= 16 + 84 + 19 + 45 \cdot 100 + 64 \cdot = 164$ 5.01 |
| 5 7 right, 3 down | 15 6 squares to the right of the line | 26 $\frac{m}{2} = 7 \cdot m = 7 \times 2 = 14$ 5.12 | 34 2 squares to the right of the line |
| 6 rotation (half turn) | 16 $\frac{1}{8}$ 4.02 | 27 a quarter turn clockwise about the dot (check each corner) | 35 a half turn: each corner the same distance on the opposite side of the dot |
| 7 reflection | 17 image 6 right (check each corner) | 28 a half turn: each corner the same distance on the opposite side of the dot | 36 $2\frac{1}{2}$ 4.04 |
| 8 0.1875 4.09 | 18 $P = 2 \times 14 + 2 \times 3 \cdot P = 34$ 5.05 | 29 a half turn: each corner the same distance on the opposite side of the dot | |
| 9 image 4 right, 2 down (check each corner) | 19 a quarter turn clockwise about the dot (check each corner) | | |
| 10 mirror image: each corner the same distance from the line, on the other side | 20 mirror image: each corner the same distance from the line, on the other side | | |
| 11 $k = 72$ 5.10 | 21 17 5.13 | | |
| 12 3 squares to the right of the line | | | |
| 13 image 4 right, 2 down (check each corner) | | | |

6.02 Combined transformations (booklet p. 5)

Examples (model these live)

E1 Translate 3 right, 1 up, then 2 right, 4 down. Write it as one translation.
across: $3 + 2 = 5$ right · up/down: 1 up, 4 down = 3 down · 5 right, 3 down

E2 Reflect the shape in the dashed line, then translate the image 2 down. Draw both.
reflect first · then move every corner 2 down

E3 A shape is turned 90° clockwise, then 180°. What single turn is that?
 $90^\circ + 180^\circ = 270^\circ$ clockwise · the same as 90° anticlockwise

Questions (25 on today's skill, 8 from earlier lessons)

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|---|---|--|--|
| 1 5 right | 12 across: $6 - 7 = -1$ · up/down: $3 - 7 = -4$ · 1 left, 4 down | 19 4 squares to the right of the line 6.01 | 26 No: moving along the mirror line gives the same result in either order. |
| 2 a full turn: back where it started | 13 first image on the other side of the line; second image 2 squares below it | 20 None is needed: the shape is back where it started. | 27 3 left and 6 down |
| 3 translation | 14 image 3 right, 1 up (check each corner) 6.01 | 21 Yes: the images end up in different places. | 28 5 left and 6 down |
| 4 9 right | 15 first image on the other side of the line; second image 2 squares below it | 22 4 left and 3 down | 29 a half turn about the dot, then that image moved 3 squares right |
| 5 270° clockwise | 16 first image 4 right; then its mirror image in the line | 23 mirror image: each corner the same distance from the line, on the other side 6.01 | 30 $9p = 216 \cdot p = 24$: 24 pencils 5.13 |
| 6 a full turn: back where it started | 17 4 right, 2 down 6.01 | 24 a half turn about the dot, then that image moved 3 squares right | 31 first image 4 right; then its mirror image in the line |
| 7 translation | 18 across: $3 - 5 = -2$ · up/down: $3 - 5 = -2$ · 2 left, 2 down | 25 $1\frac{1}{2}$ 4.04 | 32 5 right, 3 down 6.01 |
| 8 7 right | | | 33 3 left and 5 down |
| 9 $2t$ 5.07 | | | |
| 10 across: $7 - 4 = 3$ · up/down: $7 - 5 = 2$ · 3 right, 2 up | | | |
| 11 180° clockwise | | | |

6.03 Line symmetry (booklet p. 7)

Examples (model these live)

E1 How many axes of symmetry does a square have?
fold across, up and along each diagonal · 4 axes

E2 How many axes of symmetry does the letter E have?
fold across the middle · 1 axis

E3 How many axes of symmetry does a regular hexagon have?
one for every side · 6 axes

Questions (29 on today's skill, 9 from earlier lessons)

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|-----|-----|-----|-----|
| 1 4 | 2 1 | 3 7 | 4 2 |
|-----|-----|-----|-----|

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|---|---|---|--|
| 5 1 | 17 rhombus · 2 axes | 25 infinitely many (every diameter) | 31 a kite (or an isosceles trapezium) |
| 6 6 | 18 1 | 26 mirror image: each corner the same distance from the line, on the other side ^{6.01} | 32 20 axes · 10 go through opposite corners |
| 7 0 | 19 1 | 27 8 axes · 4 go through opposite corners | 33 first: $9a \cdot = 9a + 9$ ^{5.04} |
| 8 5 | 20 rotation (half turn) ^{6.02} | 28 20 ^{4.07} | 34 2 |
| 9 0 | 21 kite · 1 axis | 29 the mirror image of the half on the other side of the line | 35 translation ^{6.02} |
| 10 y ^{5.09} | 22 image 5 right, 1 up (check each corner) ^{6.01} | 30 No. A triangle has 0, 1 or 3 axes. | 36 0 |
| 11 square · 4 axes | 23 the mirror image of the half on the other side of the line | | 37 1 |
| 12 1 | 24 the mirror image of the half on the other side of the line | | 38 image 5 right, 2 down (check each corner) ^{6.01} |
| 13 rectangle · 2 axes | | | |
| 14 1 | | | |
| 15 6 squares to the right of the line ^{6.01} | | | |
| 16 2 | | | |

6.04 Rotational symmetry (booklet p. 9)

Examples (model these live)

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|---|--|--|
| E1 What is the order of rotational symmetry of a square?
fits every quarter turn · order 4 | E2 A regular pentagon: find the order and the smallest angle of turn.
order 5 · $360^\circ \div 5 = 72^\circ$ | E3 What is the order of rotational symmetry of the letter S?
fits after a half turn · order 2 |
|---|--|--|

Questions (31 on today's skill, 9 from earlier lessons)

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|--|---|---|--|
| 1 2 | 13 12 | 22 first image 3 right; then its mirror image in the line ^{6.02} | 31 a square |
| 2 1 | 14 order $24 \cdot 360^\circ \div 24 = 15^\circ$ | 23 $360^\circ \div 40^\circ \cdot =$ order 9 | 32 11 right ^{6.02} |
| 3 10 | 15 $360^\circ \div 2 \cdot = 180^\circ$ | 24 $360^\circ \div 10 \cdot = 36^\circ$ | 33 axes of symmetry: 1 · order of rotation: 1 |
| 4 1 | 16 mirror image: each corner the same distance from the line, on the other side ^{6.01} | 25 $360^\circ \div 20^\circ \cdot =$ order 18 | 34 N, S or Z |
| 5 2 | 17 axes: 4 · order: 4 | 26 an equilateral triangle | 35 axes of symmetry: 6 · order of rotation: 6 |
| 6 9 | 18 $360^\circ \div 12 \cdot = 30^\circ$ | 27 square · 4 axes ^{6.03} | 36 $\frac{t}{5} = 5 \cdot t = 5 \times 5 = 25$ ^{5.12} |
| 7 1 | 19 axes: 5 · order: 5 | 28 axes of symmetry: 2 · order of rotation: 2 | 37 $360^\circ \div 45^\circ \cdot =$ order 8 |
| 8 8 | 20 7 ^{6.03} | 29 c^2 ^{5.03} | 38 0 ^{6.03} |
| 9 5 | 21 order $15 \cdot 360^\circ \div 15 = 24^\circ$ | 30 $360^\circ \div 60^\circ \cdot =$ order 6 | 39 axes: 6 · order: 6 |
| 10 0.67 ^{4.10} | | | 40 a parallelogram |
| 11 order $4 \cdot 360^\circ \div 4 = 90^\circ$ | | | |
| 12 1 | | | |

6.05 Classifying triangles (booklet p. 11)

Examples (model these live)

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|---|--|---|
| E1 Classify a triangle with sides 5 cm, 5 cm and 7 cm.
two sides equal · isosceles | E2 Classify a triangle with angles 30° , 50° and 100° .
one angle more than 90° · obtuse-angled | E3 Classify the triangle by its sides and its angles.
two marks: two equal sides · isosceles, right-angled |
|---|--|---|

Questions (32 on today's skill, 10 from earlier lessons)

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|-----------------------------------|---|---|--|
| 1 acute-angled | 14 equilateral · acute-angled | 23 $54ax$ ^{5.08} | 34 Yes: the two shorter sides add to more than the longest. |
| 2 scalene | 15 scalene | 24 Yes: e.g. 90° , 60° , 30° . | 35 No: $4 + 4$ is not more than 8. |
| 3 at most 1 | 16 3 | 25 reflection ^{6.01} | 36 $23 - 8 - 8 \cdot = 7$ cm |
| 4 right-angled | 17 scalene · right-angled | 26 acute-angled · scalene (no equal angles) | 37 $\frac{1}{7}$ ^{4.02} |
| 5 isosceles | 18 right-angled · isosceles (two equal angles) | 27 isosceles · acute-angled | 38 No: $90^\circ + 90^\circ = 180^\circ$ leaves nothing for the third angle. |
| 6 60° | 19 mirror image: each corner the same distance from the line, on the other side ^{6.01} | 28 Yes: e.g. 120° , 30° , 30° . | 39 6 right, 1 up ^{6.01} |
| 7 acute-angled | 20 scalene · obtuse-angled | 29 Yes: the two shorter sides add to more than the longest. | 40 acute-angled · scalene (no equal angles) |
| 8 equilateral | 21 acute-angled · scalene (no equal angles) | 30 $P = 2 \times 6 + 2 \times 3 \cdot P = 18$ ^{5.05} | 41 No: $3 + 5$ is not more than 9. |
| 9 1 ^{6.04} | 22 scalene · obtuse-angled | 31 $20 - 7 - 7 \cdot = 6$ cm | 42 4 ^{6.03} |
| 10 scalene | | 32 reflection ^{6.02} | |
| 11 2 | | 33 Yes: 90° , 45° , 45° . | |
| 12 obtuse-angled | | | |
| 13 $4\frac{2}{5}$ ^{4.04} | | | |

6.06 Angle sum of a triangle (booklet p. 13)

Examples (model these live)

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|---|---|--|
| E1 Two angles of a triangle are 65° and 80° . Find the third.
$65 + 80 = 145 \cdot 180 - 145 = 35^\circ$ | E2 Find a .
$a + 50 + 70 = 180 \cdot a = 60$ | E3 An isosceles triangle has an apex angle of 40° . Find each base angle.
$180 - 40 = 140 \cdot 140 \div 2 = 70^\circ$ |
|---|---|--|

Questions (31 on today's skill, 11 from earlier lessons)

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|--|---|--|---|
| 1 56° | 15 mirror image: each corner the same distance from the line, on the other side ^{6.01} | 24 0 ^{6.03} | 33 3 squares to the right of the line ^{6.01} |
| 2 $a = 55$ | 16 $55 + 55 = 110 \cdot 180 - 110 = 70^\circ$ | 25 $70 + 70 = 140 \cdot 180 - 140 = 40^\circ$ | 34 No. $95 + 105 = 200$, which is already more than 180° . |
| 3 33° | 17 $180 - 84 = 96 \cdot 96 \div 2 = 48^\circ$ | 26 $12x = 180$, $x = 15 \cdot$ angles: $45^\circ, 60^\circ, 75^\circ$ | 35 $x + x + 10 + 106 = 180 \cdot 2x = 64 \cdot x = 32$ |
| 4 104° | 18 $75 + 75 = 150 \cdot 180 - 150 = 30^\circ$ | 27 $x + x + 40 + 56 = 180 \cdot 2x = 84 \cdot x = 42$ | 36 No. $92 + 91 = 183$, which is already more than 180° . |
| 5 $a = 40$ | 19 9 ^{4.05} | 28 HCF of 66 and 90 is 6 $\cdot = \frac{11}{15}$ ^{4.01} | 37 $\frac{1}{4}$ ^{4.03} |
| 6 57° | 20 $a + 30 + 50 = 180 \cdot a = 100$ | 29 No. $96 + 96 = 192$, which is already more than 180° . | 38 $6x = 180$, $x = 30 \cdot$ angles: $30^\circ, 60^\circ, 90^\circ$ |
| 7 70° | 21 a full turn: back where it started ^{6.02} | 30 $c = 360 - 65 - 65 \cdot c = 230$ ^{2.05} | 39 reflection ^{6.02} |
| 8 $a = 45$ | 22 $9x = 180$, $x = 20 \cdot$ angles: $20^\circ, 60^\circ, 100^\circ$ | 31 $4x = 180$, $x = 45 \cdot$ angles: $45^\circ, 45^\circ, 90^\circ$ | 40 $45 + 45 = 90 \cdot 180 - 90 = 90^\circ$ |
| 9 65° | 23 $180 - 76 = 104 \cdot 104 \div 2 = 52^\circ$ | 32 $x + x + 10 + 88 = 180 \cdot 2x = 82 \cdot x = 41$ | 41 $x + x + 30 + 108 = 180 \cdot 2x = 42 \cdot x = 21$ |
| 10 94° | | | 42 12 ^{6.03} |
| 11 10 ^{4.07} | | | |
| 12 $a + 70 + 60 = 180 \cdot a = 50$ | | | |
| 13 $a + 35 + 70 = 180 \cdot a = 75$ | | | |
| 14 $180 - 100 = 80 \cdot 80 \div 2 = 40^\circ$ | | | |

6.07 Exterior angle of a triangle (booklet p. 15)

Examples (model these live)

E1 Find x , the exterior angle.
 $x = 50 + 60 \cdot x = 110$

E2 The interior opposite angles are 35° and 75° . Find the exterior angle.
 $35 + 75 \cdot = 110^\circ$

E3 The exterior angle is 130° . One interior opposite angle is 45° . Find the other.
 $y + 45 = 130 \cdot y = 85^\circ$

Questions (30 on today's skill, 10 from earlier lessons)

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|---|---|---|---|
| 1 144° | 13 $x = 130$ | 22 a full turn: back where it started ^{6.02} | 32 interior base angle: $68^\circ \cdot$ exterior: $180 - 68 = 112^\circ$ |
| 2 $x = 125$ | 14 $x = 55 + 65 \cdot x = 120$ (exterior angle) | 23 $x + x + 15 = 81 \cdot 2x = 66 \cdot x = 33$ | 33 0 ^{6.03} |
| 3 38° | 15 $y + 35 = 110 \cdot y = 75^\circ$ | 24 $y + 75 = 160 \cdot y = 85^\circ$ | 34 interior base angle: $42^\circ \cdot$ exterior: $180 - 42 = 138^\circ$ |
| 4 79° | 16 mirror image: each corner the same distance from the line, on the other side ^{6.01} | 25 71° ^{6.06} | 35 Yes, when the interior angle next to it is obtuse. |
| 5 $x = 125$ | 17 $y + 100 = 135 \cdot y = 35$ | 26 $x + x + 5 = 43 \cdot 2x = 38 \cdot x = 19$ | 36 p ^{5.09} |
| 6 65° | 18 $y + 65 = 75 \cdot y = 10^\circ$ | 27 interior base angle: $55^\circ \cdot$ exterior: $180 - 55 = 125^\circ$ | 37 $x + x + 20 = 80 \cdot 2x = 60 \cdot x = 30$ |
| 7 136° | 19 $y + 110 = 150 \cdot y = 40$ | 28 $8 \times k \times m$ ^{5.03} | 38 translation ^{6.01} |
| 8 $x = 135$ | 20 16 ^{4.05} | 29 360° | 39 $y + 70 = 140 \cdot y = 70$ |
| 9 107° | 21 $x = 40 + 100 \cdot x = 140$ (exterior angle) | 30 10 ^{6.04} | 40 interior base angle: $51^\circ \cdot$ exterior: $180 - 51 = 129^\circ$ |
| 10 128° | | 31 $x + x + 15 = 55 \cdot 2x = 40 \cdot x = 20$ | |
| 11 $\frac{9}{11}$ ^{4.03} | | | |
| 12 $x = 65 + 70 \cdot x = 135$ (exterior angle) | | | |

6.08 Quadrilaterals (booklet p. 17)

Examples (model these live)

E1 Name the quadrilateral: 4 equal sides, no right angles.
 all sides equal $\cdot$ rhombus

E2 Name the quadrilateral and give one property.
 two pairs of parallel sides, no right angles $\cdot$ parallelogram: opposite sides equal

E3 True or false: every square is a rectangle.
 4 right angles, opposite sides equal $\cdot$ true

Questions (29 on today's skill, 10 from earlier lessons)

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|---|---|---|---|
| 1 square | 13 trapezium | 20 $\frac{1}{7}$ ^{4.02} | 30 square, rectangle, rhombus, parallelogram |
| 2 square | 14 rectangle $\cdot$ opposite sides equal, all angles 90° | 21 a rectangle or a square | 31 17° ^{6.06} |
| 3 1 | 15 True: it has 4 right angles and opposite sides equal. | 22 270° clockwise ^{6.02} | 32 a square |
| 4 rectangle | 16 mirror image: each corner the same distance from the line, on the other side ^{6.01} | 23 True: all four sides are equal. | 33 a rhombus is a parallelogram with all sides equal |
| 5 rectangle | 17 rhombus $\cdot$ all sides equal, opposite angles equal | 24 kite $\cdot$ two pairs of equal adjacent sides | 34 0 ^{6.03} |
| 6 4 | 18 False: its opposite sides are not parallel. | 25 2 ^{6.05} | 35 square, rhombus |
| 7 rhombus | 19 trapezium $\cdot$ exactly one pair of parallel sides | 26 a kite | 36 acute-angled $\cdot$ scalene (no equal angles) ^{6.05} |
| 8 parallelogram | | 27 an isosceles trapezium | 37 a square is a rhombus with right angles |
| 9 4 | | 28 a square is a special rectangle with all sides equal | 38 square, rectangle |
| 10 parallelogram | | 29 $4p = 96 \cdot p = 24$: 24 pencils ^{5.13} | |
| 11 72by ^{5.08} | | | |
| 12 square $\cdot$ all sides equal and all angles 90° | | | |

6.09 Angle sum of a quadrilateral (booklet p. 19)

Examples (model these live)

E1 Three angles of a quadrilateral are 80° , 95° and 110° . Find the fourth.
 $80 + 95 + 110 = 285$ · $360 - 285 = 75^\circ$

E2 Find a .
 $a + 70 + 90 + 115 = 360$ · $a = 85$

E3 A parallelogram has an angle of 65° . Find the other three angles.
 opposite: 65° · next to it: $180 - 65 = 115^\circ$ · 65° , 115° , 65° , 115°

Questions (32 on today's skill, 11 from earlier lessons)

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|-------------------------------|--|---|--|
| 1 30° | 15 opposite angle: 45° · next to it: $180 - 45 = 135^\circ$ | 24 opposite angle: 50° · next to it: $180 - 50 = 130^\circ$ | 34 A diagonal cuts it into two triangles: $2 \times 180^\circ = 360^\circ$. |
| 2 $a = 95$ | 16 mirror image: each corner the same distance from the line, on the other side 6.01 | 25 20° 6.06 | 35 $4x + 40 + 164 = 360$ · $4x = 156$ · $x = 39$ |
| 3 90° | 17 $2x + 170 = 360$ · $2x = 190$, $x = 95$ | 26 $10x = 360$ · $x = 36$ | 36 No. Four angles over 90° add to more than 360° . |
| 4 105° | 18 opposite angle: 145° · next to it: $180 - 145 = 35^\circ$ | 27 $4x + 30 + 166 = 360$ · $4x = 164$ · $x = 41$ | 37 $\frac{1}{2}$ 4.03 |
| 5 $a = 80$ | 19 $2x + 170 = 360$ · $2x = 190$, $x = 95$ | 28 enter each mixed numeral · $4\frac{32}{63}$ 4.09 | 38 $8x = 360$ · $x = 45$ |
| 6 180° | 20 $k - 4$ 5.04 | 29 Yes, e.g. 100° , 100° , 100° , 60° (they add to 360°). | 39 1 6.04 |
| 7 105° | 21 $a + 240 = 360$ · $a = 120$ | 30 119° 6.07 | 40 $2x + 150 = 360$ · $2x = 210$, $x = 105$ |
| 8 $a = 85$ | 22 a full turn: back where it started 6.02 | 31 $12x = 360$ · $x = 30$ | 41 $4x + 10 + 170 = 360$ · $4x = 180$ · $x = 45$ |
| 9 360° | 23 $18x = 360$ · $x = 20$ | 32 $4x + 10 + 162 = 360$ · $4x = 188$ · $x = 47$ | 42 square 6.08 |
| 10 65° | | 33 1 6.03 | 43 $6x = 360$ · $x = 60$ |
| 11 $\frac{5}{12}$ 4.12 | | | |
| 12 $a + 285 = 360$ · $a = 75$ | | | |
| 13 $a = 105$ | | | |
| 14 $a + 270 = 360$ · $a = 90$ | | | |

6.10 Properties of quadrilaterals (booklet p. 21)

Examples (model these live)

E1 A rectangle has one diagonal 13 cm long. How long is the other?
 rectangle: diagonals are equal · 13 cm

E2 A rhombus has a side of 7 cm. Find its perimeter.
 all sides equal · $4 \times 7 = 28$ cm

E3 Do the diagonals of a kite meet at right angles?
 one diagonal is an axis of symmetry · yes

Questions (36 on today's skill, 15 from earlier lessons)

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|---|--|---|---|
| 1 true | 18 opposite angles are equal · 55° | 29 equal, meet at 90° , bisect each other | 41 0 6.03 |
| 2 44 cm | 19 bisect each other | 30 opposite angles are equal · 80° | 42 the diagonals bisect each other · 5 cm and 6 cm |
| 3 4 cm | 20 mirror image: each corner the same distance from the line, on the other side 6.01 | 31 1 6.08 | 43 the diagonals bisect each other · 5 cm and 3 cm |
| 4 true | 21 opposite sides equal · $2 \times 10 + 2 \times 13 = 46$ cm | 32 a rectangle | 44 both pairs of opposite sides are parallel |
| 5 20 cm | 22 meet at 90° , one bisects the other | 33 $2x + 140 = 360$ · $2x = 220$, $x = 110$ 6.09 | 45 $a - 9$ 5.04 |
| 6 6 cm | 23 $y + 15 = 70$ · $y = 55^\circ$ 6.07 | 34 a square | 46 the diagonals bisect each other · 6 cm and 7 cm |
| 7 false | 24 meet at 90° , bisect each other | 35 the diagonals bisect each other · 6 cm and 8 cm | 47 1 6.04 |
| 8 28 cm | 25 all sides equal · $48 \div 4 = 12$ cm | 36 $= 24 + 76 + 41 + 85$ · $= 100 + 126$ · $= 226$ 5.01 | 48 it has two pairs of equal sides next to each other |
| 9 square 6.08 | 26 0.29 4.10 | 37 it has four right angles and opposite sides equal | 49 the diagonals bisect each other · 6 cm and 3 cm |
| 10 12 cm | 27 a parallelogram | 38 121° 6.07 | 50 90° 6.09 |
| 11 15 cm | 28 a full turn: back where it started 6.02 | 39 the diagonals bisect each other · 4 cm and 3 cm | 51 the diagonals bisect each other · 7 cm and 5 cm |
| 12 true | | 40 it has four equal sides | |
| 13 4 4.05 | | | |
| 14 all sides equal · $20 \div 4 = 5$ cm | | | |
| 15 16 cm | | | |
| 16 14 cm | | | |
| 17 $a = 90$ 6.09 | | | |